HACKATHON FIELD GUIDE & ARCHITECTURE WHITEPAPER

Smart Supply Chain Analyst (SSCA)

"A practical guide to how multimodal AI, graph theory, and Monte Carlo tree search predict global supply chain collapse before it happens."

Target: **Team Preparation & Judge Pitching**

Architecture: **Dual-Stack (Next.js 16 + Python MCTS)**

AI Stack: **Groq LLaMA 3.1 70B + Gemini Failover**

CHAPTER 1

The Invisible Arteries of Modern Civilization

When you buy gas, take medication, or turn on a laptop, you are touching the end of a fragile, invisible chain that stretches across thousands of miles of ocean.

Over **80% of everything the world consumes** travels on cargo vessels and oil tankers. But these ships do not roam freely across open seas—they are forced by global geography to squeeze through a handful of narrow waterways called **maritime chokepoints**:

The Strait of Malacca (Singapore): Only 1.5 miles wide at its narrowest, yet carries **90,000+ ships a year** and over 25% of all world trade.
The Strait of Hormuz: The single most critical energy bottleneck on Earth. **20.5 million barrels of crude oil** pass through it every single day.
The Suez Canal & Bab-el-Mandeb (Red Sea): The vital bridge connecting Asia and Europe.

> ⚡ **The Real-World Disaster Scenario**
>
> Imagine a drone attack strikes a tanker in the Strait of Hormuz. What happens next is a **cascading domino effect**: Tanker traffic halts → Freight insurance rates spike 400% → Crude oil cannot reach refineries in Singapore or Mumbai → Refineries throttle daily production → Petrochemical feedstock collapses → Downstream semiconductor fabs and pharmaceutical factories run dry.

CHAPTER 2

The Core Dilemma: Why Current Tools Fail

Today, trade analysts, logistics executives, and government agencies monitor supply chains using two deeply broken methods:

1. The "Dumb News Aggregator" Trap

Tools scrape hundreds of headlines and dump them into alerts. They tell you *"Conflict reported in Red Sea"*, but they cannot tell you **which specific refinery will run out of crude** or **how many days of fuel reserves remain**.

2. The "Static Spreadsheet" Trap

Risk is calculated with arbitrary, static percentages (e.g. *"Hormuz = 50% severity"*). Spreadsheets cannot model non-linear ripple effects, alternative rerouting delays, or probabilistic market escalations.

> 💡 **The Medical Analogy:**
>
> Existing tools are like a thermometer that tells a doctor "the patient has a fever".
>
> **SSCA is an MRI scan with an AI cardiologist**
>
> : it traces the exact blood clot, maps the blocked arteries, simulates 600 potential surgical outcomes, and prescribes the exact dosage of medicine before a heart attack occurs.

CHAPTER 3

The Journey of a Signal: How Information Flows

To explain our project to teammates and judges with complete confidence, follow how raw chaos in the real world is converted into a precision boardroom decision:

1

Live Multimodal Telemetry Ingestion

Sensors & APIs

Our system ingests 3 live real-time signals: (1) **Live Ship Telemetry** via WebSocket from `AISStream`, (2) **Geopolitical News** from `NewsAPI`, and (3) **Live Spot Commodity Prices** (Brent, WTI, LNG, Coal) from `EIA & API Ninjas`.

2

Sub-Second Fact Extraction (NLP)

Groq LLaMA 3.1 8B

Raw unstructured news articles are processed in under 200ms by `LLaMA 3.1 8B` running on Groq hardware. It extracts structured facts: *Which corridor? Which countries? What commodities? What is the disruption type?*

3

Knowledge Graph Traversal (Deterministic BFS)

Graph Engine

Extracted facts are anchored onto our **100+ Node Strategic Trade Knowledge Graph**. A 3-hop Breadth-First Search (BFS) traces: `Chokepoint → Affected Tankers → Raw Commodities → Dependent Refineries → Downstream Industries → Alternative Suppliers`.

4

Monte Carlo Tree Search (Probabilistic MCTS)

Python Microservice (:8787)

The graph context is dispatched to our Python FastAPI server. It plays out **600 full simulations** of the disruption across 5 sequential phases (Trigger → Triage → Rerouting → Capacity Rebuild → Resolution) to calculate the **dynamic severity %**, **escalation probability**, and **optimal counter-action**.

5

5-Agent Parallel AI Synthesis

Groq LLaMA 3.3 70B

5 specialized AI agents execute in parallel on Groq: (1) **Executive Briefing**, (2) **Supply Chain Impact**, (3) **Operational Recommendations**, (4) **Scenario Forecasting**, and (5) **Corroborated Evidence Audit Trail**.

6

Strict Schema Validation & Digital Twin Rendering

Zod & MapLibre GL 5

All AI outputs are strictly validated by `Zod` schemas. Missing data is filled from our deterministic graph, and results are projected onto the interactive 3D/2D digital twin and Bezier flow pipelines.

CHAPTER 4

Under the Hood of the Monte Carlo Engine

Why did we build a dedicated Python Monte Carlo Tree Search (MCTS) microservice instead of just asking an LLM for a percentage?

> ♟ **The Grandmaster Chess Analogy:**
> When a grandmaster plays chess, they do not just evaluate the current board—they simulate 10 moves into the future for every possible move their opponent might make.
> **Our MCTS engine does the exact same thing for global trade.**

When a crisis occurs, the future is not static—it branches into dynamic phases:

```
[ TRIGGER ] ──► [ TRIAGE ] ──► [ REROUTING ] ──► [ CAPACITY REBUILD ] ──► [ RESOLUTION ]
```

In ~2.5 milliseconds across 600 rollout simulations, the MCTS engine balances:

Exploration vs. Exploitation (UCB1 Algorithm): It tests different mitigation levers (e.g. `reserve-logistics` , `strategic-rerouting` , `spot-arbitrage` , `naval-escorts`) to find which action yields the highest economic survival score.
Dynamic Severity %: Instead of a fixed 60%, the engine might compute **54.4%** based on the likelihood of alternative maritime routing.
Escalation Risk % & Economic Stress %: Statistically computes the probability of a regional naval standoff turning into a total blockade.
Zero-Downtime Resilience: If the Python server is offline, Next.js detects it and seamlessly falls back to static baselines without throwing a single error.

CHAPTER 5

The 6 Operational Modules in Our Cockpit

1. 🌎 Geopolitical Risk & Intelligence

The "Brain". Runs 5 parallel Groq LLaMA 3.1 70B agents with an interactive "Ask Intelligence" RAG chatbot and automatic Gemini failover.

2. 🗺️ Command Center Digital Twin

The "Eyes". Interactive MapLibre GL 5 geospatial map streaming live AIS tanker fleets, chokepoint danger zones, and corridor speeds.

3. 🎮 Scenario Simulator & MCTS

The "Flight Simulator". Test disruption presets, observe 600 MCTS iterations, and flip decision levers to watch supply deficits drop.

4. 📦 Adaptive Energy Procurement

The "Wallet". Live Brent/WTI/LNG spot price tickers, supplier concentration risk (HHI index), and automated alternative vendor matching.

5. 🔬 Refinery & Processing Cockpit

The "Factory". Models refinery throughput (Jurong Island, Jamnagar), crude diet optimization (sweet vs sour), and fuel yield stress.

6. 🛢️ Strategic Reserves & Analytics

The "Shield". Simulates underground Strategic Petroleum Reserve (SPR) drawdowns, days of cover remaining, and comparative deficit charts.

CHAPTER 6

The Hackathon Pitch & Demo Script

Here is your exact 3-minute presentation playbook to wow the judges and win the hackathon:

⏱ MINUTE 1: The Visual Hook & Real-World Chaos

What to do: Open the **Command Center** with the live map. Point at the vessels moving through the Strait of Malacca and Hormuz.
What to say: *"80% of world energy flows through 4 maritime chokepoints. When a crisis happens, supply chain leaders are flying blind with static spreadsheets. We built Smart Supply Chain Analyst—the world's first multimodal digital twin and predictive disruption oracle."*

⏱ MINUTE 2: Deterministic AI Intelligence (Zero Hallucination)

What to do: Go to **Geopolitical Intelligence**. Click "Generate Intelligence Report". Show the 5 Groq agents populating in sub-2 seconds.
What to say: *"Notice how fast Groq synthesizes this report. But more importantly: our AI does not hallucinate. It is grounded in our 100-node Knowledge Graph, tracing exact dependencies from chokepoint to downstream refineries."*

⏱ MINUTE 3: The Monte Carlo Disruption Engine (The Closer)

What to do: Go to **Scenario Simulator**. Click "Hormuz Blockade". Point to the **Monte Carlo Panel**. Flip the "Release Strategic Reserves" lever.
What to say: *"This is our secret weapon. Our Python FastAPI microservice runs 600 Monte Carlo Tree Search rollouts to dynamically calculate risk. Watch what happens when we activate our emergency reserve lever—the supply deficit drops in real time."*

CHAPTER 7

How to Defend the Project to Judges (Q&A)

Judge's Question	Your Confident Technical Answer
"How do you stop LLM hallucinations?"	*"We use Knowledge Graph Grounding. A 3-hop BFS algorithm calculates the exact supply chain facts first and feeds them into Groq. The LLM only formats the briefing, and the response is validated by Zod."*
"Why use Monte Carlo MCTS?"	*"Disruptions are dynamic Markov processes with multiple branching decisions (rerouting vs escorts vs diplomatic waivers). MCTS simulates 600 potential futures to discover optimal actions that static rules miss."*
"Is this limited to one country?"	*"No, it's fully global. We have a multi-jurisdiction selector that dynamically swaps import baskets, reserve mechanisms, and port topologies (e.g. Singapore, India, Regional Hubs)."*
"What happens if an API fails?"	*"Zero downtime. We built automatic dual-LLM failover (Groq → Gemini) and fallback to deterministic baselines if the Python microservice is offline."*

🚀 Team Reminder for Demo Day:
You have a working digital twin with real WebSockets, dual-stack microservices, 5 parallel AI agents, and a Monte Carlo simulator. Speak with confidence, follow the 3-minute script, and have fun!